

You can obviously select more suitable values depending on the power rating of the adaptor, the power consumption of the device and the desired load current. You can use a mini USB connector, which makes it easy to power the circuit with contemporary mains adaptors. This also ensures that we are using a 5V supply. The maximum input current should be adjusted according to the rating of the adaptor.

You can use an NTC thermistor at CON3. Batteries are often provided with a negative temperature coefficient (NTC) thermistor, which is used to prevent them from being charged at too high or too low temperatures. The lithium-ion battery therefore has three connections: a positive terminal (BAT+), a negative terminal (BAT-) and a connection for the NTC thermistor. Note that, some batteries with three connections have only a normal resistor inside, for identification. The value of the normal resistor will be constant and will not vary with temperature of the battery.

When an NTC thermistor is used, it should be connected between THM pin and ground (via BAT- connection). A resistor (R7) is also connected between THM pin and a reference voltage (VL), which creates a potential divider. The value of the resistor is chosen such that it has the same value as the NTC thermistor at a temperature of +25°C.

The voltage at THM pin at +25°C will be equal to 0.5 VL. When the temperature rises or falls, resistance of the thermistor falls or rises, as does the voltage at THM pin. The device will charge only when this voltage is between 0.28 VL and 0.74 VL. With contemporary NTC thermistors, this corresponds to a temperature between 0°C and 50°C. When no NTC

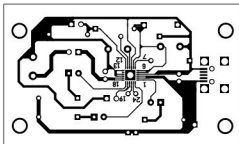


Fig. 2: Actual-size PCB layout for the battery charger

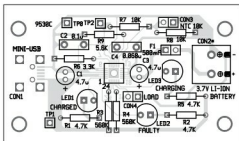


Fig. 3: Components layout of the PCB

thermistor is available, you should add R8, which causes the voltage at THM pin to be 0.5 VL. Remove R8 when NTC is available. CON4 is used for the system (load). A simple fuse (F1) offers sufficient protection from overloading.

Construction and testing

An actual-size, single-side PCB for the lithium-ion battery charger circuit is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic box. To mount SMDs you need some dexterity and soldering experience. For the TQFN-packaged MAX8677A, you should ideally use a reflow oven, since the pins and the exposed pad are on underside of the 4mmx4mm package. However, you can also solder the device by hand.

You need to make good electrical and thermal connections between the

exposed pad of the chip and the ground plane on the PCB. Mounting MAX8677A with a hot-air soldering iron is possible if you are experienced enough, though a reflow oven makes life much easier. The method described below explains how to use an ordinary soldering iron to mount the device.

Before etching the board, make sure that you make a 1.5mm dia hole in the PCB for connecting the exposed pad of the IC to the ground rail as well as providing thermal management solution to the chip. So the sequence of action for mounting the chip on the PCB should be as follows:

1. Drill a 1.5mm hole in the centre of the exposed pad
2. Position the chip on the PCB
3. Solder all contacts along the four sides of the chip. Use de-soldering wire to tidy everything up
4. Turn over the board and drop a few pieces of solder into the hole
5. Insert a solid copper wire into the 1.5mm hole and give a slight push until the wire touches the exposed pad
6. Heat the pad around the wire with soldering iron till the solder melts. When the copper wire makes good contact with the exposed pad under the chip, trim the wire with a cutter and use a file to make the protruding end flat
7. Connect the extended exposed pad to the ground plane on solder side of the PCB

You will now have a good electrical and thermal connection between the exposed pad of the chip and the ground plane. Your lithium-ion battery charger is ready for use.

Note. This circuit is based on Maxim application notes. ●



For More
Test and Equipment Stories
Visit www.electronicshobby.com